

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years** owning cloud-native, high-throughput distributed systems end to end.
- Deep in **Python and Go microservices**, event-driven architectures, real-time and batch data processing, plus **Agentic AI and RAG**.
- Track record shipping production backends, leading migrations, and improving reliability, latency, and scale.

SKILLS

- Languages** – Python, Go (Golang), SQL, C, C++.
- Backend & APIs** – FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio, Design Patterns.
- AI & Agents** – LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, pgvector.
- Data & Streaming** – Kafka, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, DynamoDB, S3.
- Cloud & Infra** – GCP, AWS, Docker, Kubernetes, Linux, CI/CD, Bazel, Observability (ELK, New Relic, Datadog, PostHog, LangSmith).
- Core** – Distributed Systems, Microservices, HLD/LLD, Fault Tolerance, Concurrency, Caching, Sharding, Mentorship.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart (AI-Powered Retail Merchandise Planning)

- Building **AssortSmart** (what / how much / which stores) with **FastAPI, LangGraph, MCP + Go (Gin)**; targeting **under 1h** and **≥20 configs/plan**.
- Cutting clustering failures **8.5% (37/437 kik)** toward **under 2%** via **14 R/O tools** and **3 confirm gates**; agent never writes SQL.
- Replacing a projected **12B-row** design with **~25M aggregates** — sub-second rollups, **~0.4 ms** cell edits (measured).
- Adopting **ClickHouse (63 tables / 8 layers, agent R/O)**; POC cut **250M-row pivots 189s → 12.3s (~15.5x)**.
- Tech** – Python, FastAPI, LangGraph, MCP, Go, Gin, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, GCP, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owned **FRM Risk Scoping (FastAPI, MySQL, React)**: **8 screens, 30+ APIs, \$340M** materiality; target **70%** recon cut (**~2 weeks → 3–4 days**).
- Designed an **11-table SQLAlchemy 2.0 MySQL** schema auto-flagging material items across **55 line items** and **14 entities**.
- Owned Sheets → **MySQL v2 recon** across **18 files**; validated HFM vs **10-Q**; fixed a column-aliasing bug.
- Led 3 engineers** through design reviews, API contracts, and CI gates for PwC audit releases.
- Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Cut menu onboarding **24h → 2h** and saved **\$600K+/yr** on **30K+ menus/month** with **Selenium** scrapers on GCP.
- Hit **98% fidelity / 100% schema consistency** (offline) via **RAG + Gemini 2.5 Pro** extraction and SFT.
- Raised successful ingestions to **95%+** via IP rotation, dynamic proxies, and adaptive backoff.
- Automated **ANZ driver doc compliance to 99.9%**, saving **20 hrs/week**.
- Tech** – Python, Selenium, Gemini, RAG, SFT, GCP, Docker.

Software Development Engineer 2 MASTERS INDIA, Noida December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Led PHP → **FastAPI** strangler for **1,500+ clients**; cut p95 **1.2s → 300ms**; mentored **2 engineers**.
- Built Kafka + PG quarter-shard e-invoicing: **1M+ IRP/day, 100K+/import**; lifted throughput **700 → 4,000 req/min**.
- Hardened IRP with **idempotency keys, retries, DLQ**; cut redundant DB reads **30%** with Redis.
- Cut incident triage **70%** with **ELK / New Relic**; coverage **35% → 82%** at **98%** deploy success.
- Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer GEEKSFORGEEKS, Noida August 2021 – November 2022

- Improved reliability for **10K+ daily queries** and **10x** contest spikes via PHP → **Django** migration.
- Lifted premium subscriptions **15–20%** with voting/pinning APIs on MySQL, MongoDB, Redis, Elasticsearch.
- Raised course sales **30%** and ops efficiency **70%** via influencer analytics and video-processing crons.

EDUCATION

B.Tech (Information Technology) Institute of Engineering & Technology, Lucknow July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in Google Code Jam Qualifier 2021; Smart India Hackathon 2020 finalist (top 3 nationally, MHRD).
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certs ([HackerRank](#), [LangChain](#)).